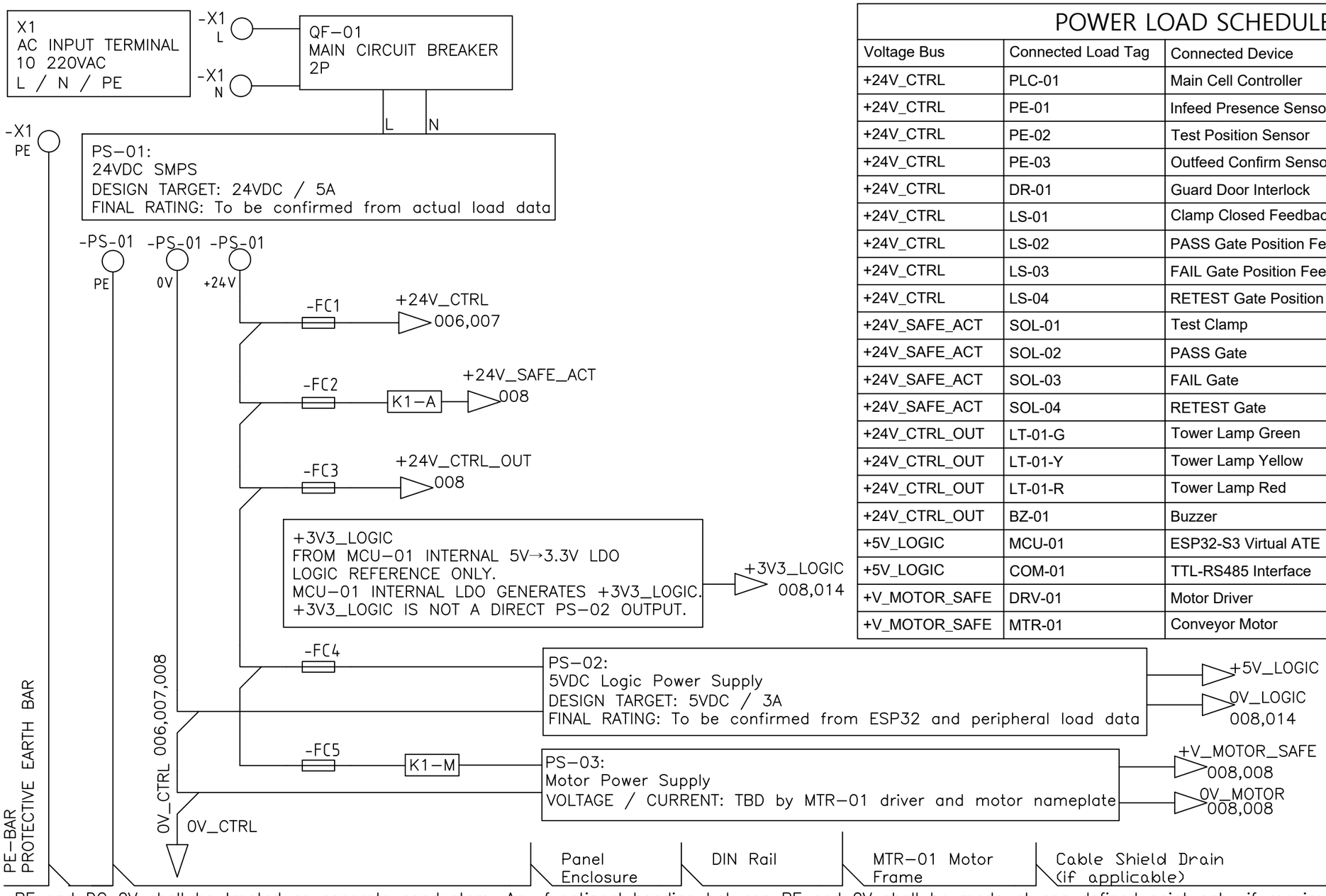


PE and DC 0V shall be treated as separate conductors. Any functional bonding between PE and 0V shall be made at one defined point only, if required by the final design.

AC INPUT
1Ø 220VAC, L / N / PE
External supply rating
to be verified before fabrication.

K1-A and K1-M are safety-controlled contacts
operated by the same safety circuit.
REF: DRAWING 006 SAFETY CIRCUIT

Fuse ratings and characteristics for FU-01 through FU-05
shall be finalized based on actual load current,
inrush current, conductor rating, and component datasheets.

POWER LOAD SCHEDULE

Voltage Bus	Connected Load Tag	Connected Device	Reference
+24V_CTRL	PLC-01	Main Cell Controller	DWG 004
+24V_CTRL	PE-01	Infeed Presence Sensor	DWG 007
+24V_CTRL	PE-02	Test Position Sensor	DWG 007
+24V_CTRL	PE-03	Outfeed Confirm Sensor	DWG 007
+24V_CTRL	DR-01	Guard Door Interlock	DWG 006 / 007
+24V_CTRL	LS-01	Clamp Closed Feedback	DWG 007
+24V_CTRL	LS-02	PASS Gate Position Feedback	DWG 007
+24V_CTRL	LS-03	FAIL Gate Position Feedback	DWG 007
+24V_CTRL	LS-04	RETEST Gate Position Feedback	DWG 007
+24V_SAFE_ACT	SOL-01	Test Clamp	DWG 008
+24V_SAFE_ACT	SOL-02	PASS Gate	DWG 008
+24V_SAFE_ACT	SOL-03	FAIL Gate	DWG 008
+24V_SAFE_ACT	SOL-04	RETEST Gate	DWG 008
+24V_CTRL_OUT	LT-01-G	Tower Lamp Green	DWG 008
+24V_CTRL_OUT	LT-01-Y	Tower Lamp Yellow	DWG 008
+24V_CTRL_OUT	LT-01-R	Tower Lamp Red	DWG 008
+24V_CTRL_OUT	BZ-01	Buzzer	DWG 008
+5V_LOGIC	MCU-01	ESP32-S3 Virtual ATE	DWG 003
+5V_LOGIC	COM-01	TTL-RS485 Interface	DWG 003
+V_MOTOR_SAFE	DRV-01	Motor Driver	DWG 008
+V_MOTOR_SAFE	MTR-01	Conveyor Motor	DWG 008

POWER DISTRIBUTION SUMMARY

Branch	Source	Protection	Voltage Bus	Load Group	E-Stop Behavior
BR-01	PS-01	FU-01	+24V_CTRL	PLC / Sensors / Communication	Maintained
BR-02	PS-01	FU-02 + K1-A	+24V_SAFE_ACT	Clamp / Bin Gates	Removed
BR-03	PS-01	FU-03	+24V_CTRL_OUT	Tower Lamp / Buzzer	Maintained for alarm indication
BR-04	PS-01	FU-04 + PS-02	+5V_LOGIC	MCU / Communication Logic	Maintained
BR-05	PS-01	FU-05 + K1-M + PS-03	+V_MOTOR_SAFE	Motor Driver / Conveyor Motor	Removed
BR-06	X1-PE	PE-BAR	PE	Protective Earth	Always connected

[GENERAL POWER DESIGN NOTES]
[1] Final breaker, fuse, SMPS, relay, and conductor ratings shall be selected using the actual component datasheets and measured load current before fabrication.
[2] PLC-01, MCU-01, and communication power shall remain energized during E-Stop for alarm monitoring and diagnostics.
[3] Actuator and motor power shall be removed by a hardwired safety-controlled disconnect device during E-Stop.
[4] PLC outputs shall not directly supply motor power. Interposing relays, drivers, or contactors shall be used.
[5] PE, 0V_CTRL, 0V_LOGIC, and 0V_MOTOR shall be managed as separate conductors. Any bonding point shall be deliberate and documented.
[6] All primary DC distribution branches shall have individual protection devices appropriate to the final connected load.

NO.	DATE	REVISION
A	2026-08-17	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell
Power Single-Line Diagram

DRAWN BY: J. H. KO
DATE: 2026-08-17
DWG NO: 005
SHEET NO: 1 OF 1